

Reviewer Pack — Minimal Rank of Symplectic Forms over Boolean Algebras vs Re...

Entry c43dbad57206 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Symplectic Forms over Boolean Algebras vs Resolution Proof Width for Random k-CNF

Entry ID: c43dbad57206 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 03:15:36 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity (for random k-CNF)

Statement:

[‘For a random k-CNF formula F with n variables and m clauses, the minimal rank of the symplectic form associated with F is at most $c\sqrt{m/n}$, where c is a constant.’, ‘For any instance F of size $n \leq 40$, there exists an integer r such that the resolution proof width for F is at least rn .’, ‘The correlation coefficient between the minimal rank of the symplectic form and the resolution proof width for instances of size $n \leq 40$ is greater than 0.7.’]

Rationale (proposer’s reasoning):

[‘Symplectic forms are known to have a rich algebraic structure that could potentially encode non-trivial combinatorial properties.’, ‘Random k-CNF formulas provide a natural setting to study complexity theory, where symplectic forms might reveal structural properties related to resolution proof complexity.’, ‘By studying the relationship between symplectic geometry and resolution proof complexity, we may uncover new insights into the nature of computational problems.’]

Taxonomy category: Symplectic_Geometry_x_Complexity_Theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 95b91c8deb62b233

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for instances of size $n \leq 40$, the mean minimal rank of symplectic forms is at most $c\sqrt{m/n}$ AND the mean resolution proof width is at least rn , where c and r are constants. It is falsified if either the mean minimal rank exceeds $c\sqrt{m/n}$ or the mean resolution proof width is less than rn .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - symplectic geometry AND resolution proof complexity - Boolean algebra AND minimal rank of symplectic forms - random k-CNF AND resolution proof width AND symplectic form

Top relevant hits considered: - [<http://arxiv.org/abs/1612.04764v1>] Cohomological aspects on complex and symplectic manifolds - [<http://arxiv.org/abs/1811.09984v4>] Translated points for contactomorphisms of prequantization spaces over monotone symplectic toric manifolds - [<http://arxiv.org/abs/1002.3099v3>] Tamed Symplectic forms and SKT metrics - [<http://arxiv.org/abs/math/0405176v4>] Quantized symplectic oscillator algebras of rank one - [<http://arxiv.org/abs/1307.7151v2>] Root Systems In Finite Symplectic Vector Spaces - [<http://arxiv.org/abs/1803.11111v1>] Higher symmetries of symplectic Dirac operator - [<http://arxiv.org/abs/2301.10131v2>] Random perfect matchings in regular graphs - [<http://arxiv.org/abs/1510.02335v3>] Cliques in dense inhomogeneous random graphs

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_cnf(n, m):
        clauses = []
        for _ in range(m):
            clause = set(random.sample(range(1, n + 1), k=random.randint(1, n)))
            if random.choice([True, False]):
                clause = {-(var) for var in clause}
            clauses.append(clause)
        return clauses

    def symplectic_form(clauses):
        n = len(clauses[0])
        form = [[0] * (2 * n) for _ in range(2 * n)]
        for clause in clauses:
            for lit in clause:
```

```

        if lit > 0:
            i, j = lit - 1, lit - 1
        else:
            i, j = -(lit + 1), -(lit + 1)
        form[i][j] += 1
        form[j][i] += 1
    return form

def resolution_width(clauses):
    # Simplified version of resolution width calculation
    # This is a placeholder and should be replaced with an actual algorithm
    return len(clauses) * 2

n = random.randint(10, 40)
m = random.randint(n, n * 3)
clauses = generate_k_cnf(n, m)

form = symplectic_form(clauses)
width = resolution_width(clauses)

# Placeholder for minimal rank calculation
min_rank = sum(max(form[i][j] for j in range(2 * n)) for i in range(2 * n))

return {
    "metric_name": "min_rank",
    "metric_value": min_rank,
    "instances_tested": 1,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else list(range(2, 30))

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d15e4a1b.py", line 72, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d15e4a1b.py", line 56, in run_trial
    min_rank = sum(max(form[i][j] for j in range(2 * n)) for i in range(2 * n))
                  ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d15e4a1b.py", line 56, in <genexpr>
    min_rank = sum(max(form[i][j] for j in range(2 * n)) for i in range(2 * n))
                  ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d15e4a1b.py", line 56, in <genexpr>
    min_rank = sum(max(form[i][j] for j in range(2 * n)) for i in range(2 * n))
                  ~~~~~
```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that the pre-registered support conditions could not be verified. | next: Re-run the test to ensure it completes successfully and produces the required data for analysis.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11998
2	preregistration	ollama_remote	glm4:latest	0	0	5940
3	novelty	ollama_remote	glm4:latest	0	0	4510
4	novelty	ollama_remote	glm4:latest	0	0	6557
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16994
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12417
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9474
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7840
9	judge	ollama_remote	glm4:latest	0	0	8246

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 83976 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/c43dbad57206.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c43dbad57206.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/c43dbad57206.tar.gz` (if generated)